

Unit 35: Inverse Trigonometric Functions — Full Solutions

Warm-up

1. $\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$

2. $\arccos(0) = \frac{\pi}{2}$

3. $\arctan(1) = \frac{\pi}{4}$

4. $\arcsin\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\pi}{4}$

5. $\arccos(1) = 0$

6. $\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1-x^2}}$

7. $\frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$

Standard

8. $\sin\left(\arccos\left(\frac{3}{5}\right)\right)$

Let $\theta = \arccos\left(\frac{3}{5}\right)$, so $\cos\theta = \frac{3}{5} = \frac{\text{adjacent}}{\text{hypotenuse}}$. Draw a right triangle with adjacent = 3, hypotenuse = 5. By the Pythagorean theorem, opposite = $\sqrt{25-9} = \sqrt{16} = 4$.

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{4}{5}$$

Answer: $\frac{4}{5}$.

9. $\cos\left(\arcsin\left(\frac{5}{13}\right)\right)$

Let $\theta = \arcsin\left(\frac{5}{13}\right)$, so $\sin\theta = \frac{5}{13} = \frac{\text{opposite}}{\text{hypotenuse}}$. Opposite = 5, hypotenuse = 13. Adjacent = $\sqrt{169-25} = \sqrt{144} = 12$.

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{12}{13}$$

Answer: $\frac{12}{13}$.

10. $\tan\left(\arcsin\left(\frac{1}{3}\right)\right)$

Let $\theta = \arcsin\left(\frac{1}{3}\right)$: opposite = 1, hypotenuse = 3. Adjacent = $\sqrt{9-1} = \sqrt{8} = 2\sqrt{2}$.

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$$

Answer: $\frac{\sqrt{2}}{4}$.

11. $\frac{d}{dx}[\arcsin(2x)]$

Chain rule: $u = 2x$, $u' = 2$.

$$\frac{d}{dx}[\arcsin(2x)] = \frac{2}{\sqrt{1-4x^2}}$$

12. $\frac{d}{dx}[\arctan(x^2)]$

Chain rule: $u = x^2$, $u' = 2x$.

$$\frac{d}{dx}[\arctan(x^2)] = \frac{2x}{1+x^4}$$

13. $\frac{d}{dx}[\arccos(3x)]$

Chain rule: $u = 3x$, $u' = 3$.

$$\frac{d}{dx}[\arccos(3x)] = \frac{-3}{\sqrt{1-9x^2}}$$

14. $\sec(\arctan(2))$

Let $\theta = \arctan(2)$: opposite = 2, adjacent = 1, hypotenuse = $\sqrt{4+1} = \sqrt{5}$.

$$\sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{\sqrt{5}}{1} = \sqrt{5}$$

Answer: $\sqrt{5}$.

Challenge

15. $y = \arctan(\sin x)$. Find $\frac{dy}{dx}$.

Chain rule: $u = \sin x$, $u' = \cos x$.

$$\frac{dy}{dx} = \frac{\cos x}{1+\sin^2 x}$$

16. $\sin\left(\arccos\left(\frac{\sqrt{2}}{2}\right)\right)$

Recognize $\frac{\sqrt{2}}{2}$ as a standard cosine value: $\arccos\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4}$ (since $\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$).

$$\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

Answer: $\frac{\sqrt{2}}{2}$.

17. $\tan\left(\arcsin\left(-\frac{1}{2}\right)\right)$

Recognize $-\frac{1}{2}$ as a standard sine value: $\arcsin\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$ (since $\sin\left(-\frac{\pi}{6}\right) = -\frac{1}{2}$, and this angle lies within $\arcsin$'s range).

$$\tan\left(-\frac{\pi}{6}\right) = -\tan\left(\frac{\pi}{6}\right) = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$$

Answer: $-\frac{\sqrt{3}}{3}$.

18. $\frac{d}{dx}\left[x \arcsin(x) + \sqrt{1-x^2}\right]$

First term ($x \arcsin x$), product rule: $f = x$ ($f' = 1$), $g = \arcsin x$ ($g' = \frac{1}{\sqrt{1-x^2}}$).

$$\frac{d}{dx}[x \arcsin x] = \arcsin x + \frac{x}{\sqrt{1-x^2}}$$

Second term ($\sqrt{1-x^2} = (1-x^2)^{1/2}$), chain rule: inner $u = 1-x^2$ ($u' = -2x$).

$$\frac{d}{dx}[(1-x^2)^{1/2}] = \frac{1}{2}(1-x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{1-x^2}}$$

Combine both terms:

$$\left(\arcsin x + \frac{x}{\sqrt{1-x^2}}\right) + \left(\frac{-x}{\sqrt{1-x^2}}\right) = \arcsin x$$

The two fraction terms cancel exactly.

$$\frac{d}{dx}\left[x \arcsin(x) + \sqrt{1-x^2}\right] = \arcsin x.$$

19. $\cos\left(\arctan\left(\frac{3}{4}\right)\right)$

Let $\theta = \arctan\left(\frac{3}{4}\right)$: opposite = 3, adjacent = 4, hypotenuse = $\sqrt{9+16} = \sqrt{25} = 5$.

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{4}{5}$$

Answer: $\frac{4}{5}$.

Applied

20. $\theta = \arctan\left(\frac{15}{8}\right)$

Note that 8-15-17 is a Pythagorean triple ($8^2 + 15^2 = 64 + 225 = 289 = 17^2$), so this angle isn't one of the standard "nice" angles, but we can still evaluate it numerically:

$$\theta = \arctan(1.875) \approx 1.081 \text{ radians} \approx 61.93^\circ$$

Answer: $\theta \approx 61.93^\circ$.

21. $\theta(t) = \arctan\left(\frac{x(t)}{50}\right)$, with $x = 30$, $\frac{dx}{dt} = 10$.

Using the chain rule (treating $\frac{x}{50}$ as the "u" inside arctan):

$$\frac{d\theta}{dt} = \frac{1}{1 + \left(\frac{x}{50}\right)^2} \cdot \frac{1}{50} \cdot \frac{dx}{dt}$$

Plug in $x = 30$: $\frac{x}{50} = \frac{30}{50} = 0.6$, so $1 + (0.6)^2 = 1 + 0.36 = 1.36$.

$$\frac{d\theta}{dt} = \frac{1}{1.36} \cdot \frac{1}{50} \cdot 10 = \frac{1}{1.36} \cdot \frac{10}{50} = \frac{1}{1.36} \cdot 0.2$$

Working with exact fractions instead: $\frac{x}{50} = \frac{3}{5}$, so $1 + \left(\frac{3}{5}\right)^2 = 1 + \frac{9}{25} = \frac{34}{25}$.

$$\frac{d\theta}{dt} = \frac{1}{\frac{34}{25}} \cdot \frac{1}{50} \cdot 10 = \frac{25}{34} \cdot \frac{10}{50} = \frac{25}{34} \cdot \frac{1}{5} = \frac{5}{34}$$

Answer: $\frac{d\theta}{dt} = \frac{5}{34} \approx 0.147 \text{ rad/s}$.

22. $\theta = \arcsin\left(\frac{x}{10}\right)$

Chain rule: $u = \frac{x}{10}$, $u' = \frac{1}{10}$.

$$\frac{d\theta}{dx} = \frac{1/10}{\sqrt{1 - \left(\frac{x}{10}\right)^2}} = \frac{1}{10\sqrt{1 - \frac{x^2}{100}}} = \frac{1}{\sqrt{100 - x^2}}$$

(simplifying by pulling the 10 inside the square root: $10\sqrt{\frac{100-x^2}{100}} = \sqrt{100-x^2}$).

At $x = 6$:

$$\left. \frac{d\theta}{dx} \right|_{x=6} = \frac{1}{\sqrt{100-36}} = \frac{1}{\sqrt{64}} = \frac{1}{8}$$

Answer: $\frac{d\theta}{dx} = \frac{1}{\sqrt{100-x^2}}$, and at $x = 6$, this equals $\frac{1}{8}$.

23. $\theta = \arccos\left(\frac{3}{5}\right)$

This is the same triangle as Problem 8 (a 3-4-5 triangle):

$$\theta = \arccos(0.6) \approx 53.13^\circ$$

Answer: $\theta \approx 53.13^\circ$.